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ME315 Final Project:
*Predicting Track Popularity & Genre
from Spotify Audio Features*

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1. Introduction

Music streaming platforms characterize every track by a set of algorithmically derived audio features such as acousticness, danceability, and energy. These features expose two natural supervised-learning questions. First, to what extent can a track's popularity be predicted from its audio features alone (a regression task)? Second, can a track's genre be determined from the same features (a multi-class classification task)? The two tasks are intentionally distinct in type, fulfilling the course requirement to address two different machine learning problems.

The genre task was chosen after an initial classification target, musical mode (major vs. minor key), was analyzed and found to be near-orthogonal to the audio features; every feature-mode correlation was below 0.07 in absolute value, and the largest standardized class-mean difference was only 0.13, so no model could be expected to exceed the majority-class rate. On the other hand, genre is strongly expressed in the audio features, making it a target for which competing models can be meaningfully compared. The aim throughout this project is not a single model, but a comparison of methods for each task (regression and classification), under a consistent cross-validation and hold-out protocol, with attention to explainability and to the limitations of the feature set.

1.1 Data Description

We used the "Ultimate Spotify Tracks DB" dataset (Kaggle, user zaheenhamidani), supplied as SpotifyFeatures.csv. The raw dataset included 232,725 records. However, some tracks appeared multiple times because they were classified under multiple genres, leading to duplicate records of the same track. We created two new datasets for our regression and classification tasks. For the regression task, we first eliminated all duplicate records based on track_id. We also eliminated one record which was missing its track name. After eliminating both duplicates and records with missing track names, we were left with 176,773 unique tracks. For the classification task, we selected six genres that were acoustically distinct from each other and had sufficient records to support our analysis (Classical, Comedy, Country, Electronic, Hip-Hop, Jazz). Because the raw duplication allows for a track to be associated with multiple genres, we then re-extracted the subset of tracks that fall exclusively into only one of the six genres. As such, we dropped approximately 1.5% of the candidate tracks that were associated with more than one of the six genres.

Ultimately, this resulted in 54,070 tracks that were relatively evenly distributed among the six genres (ranging from 8,603 tracks to 9,680 tracks per genre; a difference of 1.13x between the largest and smallest number of tracks). Both tasks use the same twelve predictors: acousticness, danceability, energy, instrumentalness, liveness, loudness, speechiness, tempo, valence, duration_min (duration converted from milliseconds to minutes), mode_enc (an indicator for major key), and is_4_4 (an indicator for a 4/4 time signature). Identifier and free-text columns (track_id, artist_name, track_name) are dropped as they are not usable numeric predictors. Popularity is the regression target; genre is the classification target and is not itself a predictor.

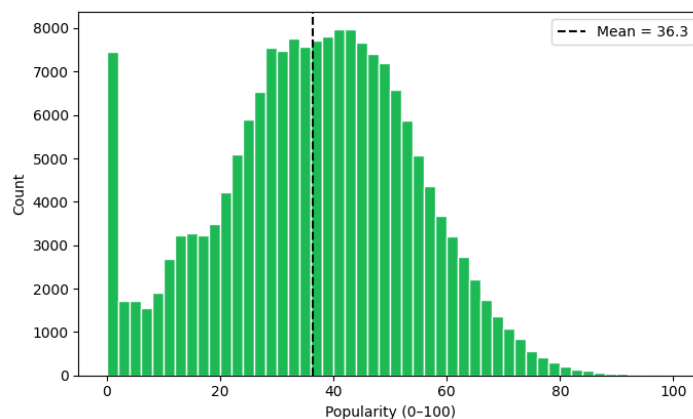
2. Exploratory Data Analysis

2.1 Target Distributions

Figure 1 shows the regression target: “popularity.” Popularity is a somewhat bell-shaped, slightly right-skewed variable having an average of 36.3 and a standard deviation of 17.4. It has a range from 0 to 100. There is a small spike at exactly zero, however it accounts for less than 3.5% of tracks, therefore there is no need to perform any type of transformation on the variable. The standard deviation (17.4) is very similar to the test RMSE of a baseline that predicts the mean for every track (17.42), so it serves as the reference point for the regression results.

The classification variable “genre” has been designed to have a relatively similar number of tracks within each of the six retained genres; tracks range in count from Country (8603 tracks) to Comedy (9680 tracks). Therefore, if a classifier were to predict the most common genre (Comedy) all the time it would achieve an accuracy of 0.179. This is considered the majority-class baseline for the model to beat. The near equality of counts among the genres eliminates the need for resampling or weighting by genre and allows for standard interpretation of multi-class metrics.

Figure 1: Popularity distributions



2.2 Feature Distributions & Outliers

Figure 2 depicts the ten continuous features that were calculated from the entire deduplicated dataset. The continuous features have three types of distributions. Some of the features appear to be relatively symmetrical and normally distributed (danceability, energy, tempo, valence; all of these features have $|\text{skew}|$ values less than 0.5). Other features are heavily skewed to the right, having a high density of data points around zero; i.e., instrumentalness, speechiness, and liveness tend to cluster at lower values due to most songs being vocal, non-spoken, and studio recorded. Loudness is negatively skewed (measured in decibels, typically ranging from -20dB to 0dB), whereas acousticness is bimodal. Specifically, tracks tend to be either highly acoustic or highly non-acoustic. Regarding the two binary flags, the 4/4 time signature indicator is highly prevalent at 84.4%, thus representing a low variance predictor.

Figure 2: Distributions of the ten continuous audio features

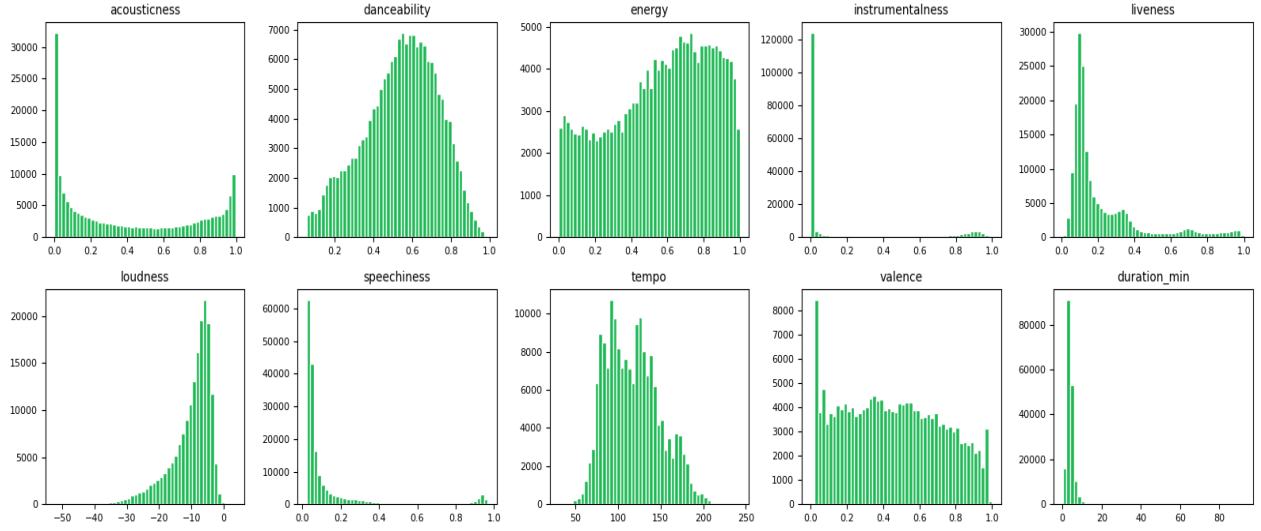


Table 1 lists the skewness and number of outliers, based on $1.5 \times \text{IQR}$, of each of the continuous features. The large outlier percentages for those features that have many pile-ups at their lower boundaries (instrumentalness 21.9%, speechiness 14.5%, liveness 10.0%) are simply an artifact of applying a symmetrical rule to a heavily one-sided distribution. A spoken word track with a high score for speechiness is a real observation, not a data error. Thus, we will keep all rows and not remove any of these values. One feature that requires further analysis is `duration_min`. Its skewness of 9.45 indicates a very long tail of extended tracks. Because the tree-based models we use later do not depend upon scale or skewness, we will not transform this variable. However, if we were using linear models, a log transformation could be a viable option.

Table 1: Skewness and IQR-based outlier counts for the ten continuous features (full dataset, $n = 176,773$).

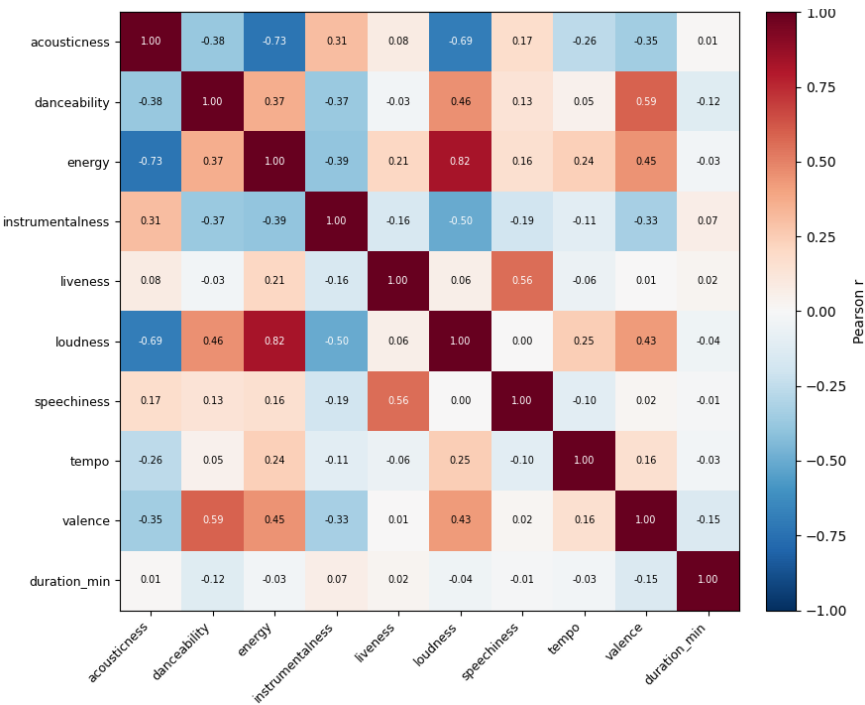
Feature	Skew	Outliers (n)	Outliers (%)
acousticness	0.37	0	0.0%
danceability	-0.33	0	0.0%
energy	-0.33	0	0.0%
instrumentalness	1.58	38,724	21.9%
liveness	1.96	17,671	10.0%
loudness	-1.48	8,621	4.9%
speechiness	3.10	25,715	14.5%
tempo	0.41	107	0.1%
valence	0.15	0	0.0%
duration_min	9.45	10,526	6.0%

2.3 Correlation Structure

The correlation matrix of the numerical variables, shown in Figure 3 (next page), indicates the strongest association among them was between Energy and Loudness ($r = 0.83$). The next strongest associations were Energy-Acousticness ($r = -0.73$) and Loudness-Acousticness ($r = -0.69$). It seems as if louder, more

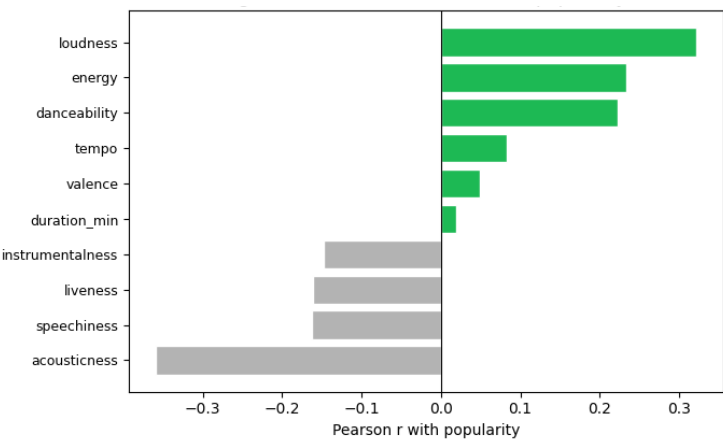
energetically performed songs are less acoustic, which was an expected outcome. While these associations are moderate to strong, they are still far from being collinear ($|r| > .95$) enough to cause problems with ordinary least squares estimation. Therefore, while there may be some degree of redundancy present in the data set, it is unlikely that using regularization techniques will substantially reduce the variance of the estimators produced by ordinary least squares.

Figure 3: Correlation matrix of continuous audio features



In Figure 4, we observe how strongly each individual feature relates to popularity. The three most strongly positively related features include: Loudness ($r = 0.32$), Energy ($r = 0.23$) and Danceability ($r = 0.22$); whereas Acousticness is the single strongest negatively related feature ($r = -0.36$). While Valence is positively but very poorly related to popularity at the marginal level ($r = 0.05$), it will be interesting to see if the relationship changes sign or strengthens once other features are controlled for.

Figure 4: Feature correlations with popularity

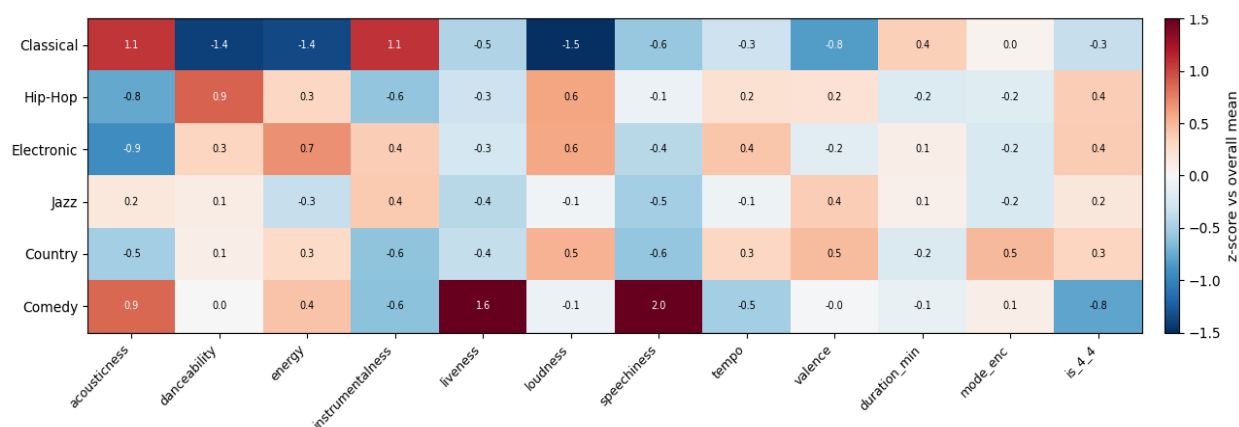


2.4 Class-conditional Feature Behavior

Figure 5 represents the average of each feature per genre (a genre signature), based upon the classification data set. Each genre has a unique signature that is both musically cohesive and distinctly different. As an example, comedy is very high in terms of speechiness (+2.0 standard deviations) and liveness (+1.6), indicating that many comedy recordings include spoken words and live audiences.

In contrast, classical music is high in terms of acousticness and instrumentalness (both +1.1), while being relatively low in terms of loudness (-1.5), danceability (-1.4), and energy (-1.4). This is nearly the exact opposite of electronic music (high energy, low acousticness). Hip-Hop has a signature characterized by high levels of danceability and low acousticness. Jazz has the least distinct signature of all six genres, as there is not one particularly strong characteristic of jazz; therefore, it will likely be difficult to classify jazz, and it may frequently be mistaken for other acoustically similar genres.

Figure 5: Per-genre feature signatures (standardized means)



The results from the analysis of the features for predicting genre can be seen in Table 2, where we report the eta-squared value for each of our features. The eta-squared represents how much of the variation in the feature can be attributed to the genre variable. As you can see, genre explains a large share of the variance in speechiness (0.86), acousticness (0.62), liveness (0.59), and loudness (0.55). Conversely, mode_enc (0.06) and duration_min (0.04) represent very little of the variance in their respective features. In comparison to the abandoned mode variable, this difference is quite apparent; none of the features were able to explain a substantial amount of the variance in mode, while many of the same features were able to explain a considerable amount of the variance in genre. Therefore, it is evident that audio features can be used to meaningfully predict genre.

Table 2: Feature discriminative power for genre (eta-squared), classification subset ($n = 54,070$).

Feature	eta-squared (variance explained by genre)
speechiness	0.86
acousticness	0.62
liveness	0.59
loudness	0.55
danceability	0.49
energy	0.47
instrumentalness	0.43
is_4_4	0.20
valence	0.19
tempo	0.12
mode_enc	0.06
duration_min	0.04

LDA assumes a shared covariance matrix across classes; because this assumption holds well in our data (relative Frobenius differences of only 0.02–0.18) and LDA outperformed QDA in cross-validation (0.743 vs. 0.733), we used LDA as our main discriminant method.

3. Methods

3.1 Workflow & Pre-processing

Data from each of the tasks were divided into both a training and testing set using a common random number generator (seed) to ensure that results can be replicated. Continuous variables were standardized (i.e. had their average value removed and then divided by their standard deviation); this process was carried out only on the training data, as part of a pipeline, so that no information about the testing data could influence how standardization was performed. The regularization parameters used in the linear model were determined through five-fold cross-validation of the training data; the tree-based models used reasonable defaults that were designed to limit overfitting. The final performance of each model was evaluated on the test data. The standardization of continuous variables is necessary for the linear and discriminant models due to large differences in scale among many of the input variables (e.g., decibels vs. 0-1 audio descriptors), whereas the tree-based models do not depend on the scale of the inputs.

3.2 Regression Models (Task 1)

We have used five models from two families. Three are linear; these include OLS as a baseline model, and two regularized models: Ridge regression and Lasso regression, with penalty strengths tuned by 5-fold cross-validation. The other two are nonlinear ensemble methods based on trees: Random Forest (bagging), and XGBoost (gradient boosting). The purpose of including the two nonlinear models was to determine if the low R^2 for linear models is due to an upper bound in the data itself or if it is just a shortcoming of linear models. All of the models were fit by minimizing the square of errors.

3.3 Classification Models (Task 2)

The four classifier comparisons are between multinomial logistic regression (with tuning for C , the inverse regularization parameter), linear discriminant analysis (LDA) which was used as the primary discriminant, quadratic discriminant analysis (QDA) that is presented with LDA, and a Random Forest to provide an additional means of capturing non-linear structure. The LDA classifier has been chosen as the primary discriminant based on the results from both cross-validation and covariance analysis in Section 2.4. Each model predicts one of the six genre classes, and the train/test split maintains the class distribution of the subset.

3.4 Evaluation Metrics

The regression models were evaluated using the root-mean-square error (RMSE) and the R-squared value on a test set. The results were compared to those obtained from the mean-prediction baseline. Classification models were evaluated using accuracy relative to the 0.179 majority class baseline; as well as macro-average precision, recall and F1-score (which is appropriate due to the nearly balanced nature of the multi-class problem); and a confusion matrix for the best model to show which genre types are confused with other types.

4. Results

4.1 Regression

Table 3 compares all the models we tested. Our baseline is predicting the average popularity of each song. This results in a test RMSE of 17.42 and an R^2 value of near zero. All our other models need to outperform this reference point. All three linear models have almost identical test RMSE values (15.57) and test R^2 values (0.201). When we use cross-validation to select regularization parameters for our linear models, we find that a Ridge penalty of 50 and a Lasso penalty of 0.001 are optimal. It is not surprising that regularization did not improve our unpenalized OLS model. With approximately 141,000 training samples and only 12 features, the ordinary least squares estimator has very little variability, so there is no room for shrinkage to decrease variability. Our two nonlinear ensemble models provide us with some additional insights. Random Forest and XGBoost both produce test R^2 values that are substantially higher than those produced by our linear models (33.9% and 34.2% respectively). These values represent an approximately 14 percentage point increase in R^2 over the linear models. XGBoost produces a slightly better overall performance than Random Forest; however, they perform similarly enough that they could be interchangeable in practice. The increased performance demonstrates that popularity is affected by the features of the songs in ways that a linear model cannot account for.

Specifically, the features interact and create thresholds that require a more flexible model to accurately represent them. We also used regularization to prevent overfitting in our ensemble models. For Random Forest, we required each leaf to contain at least 30 observations. For XGBoost, we limited the depth of each tree to 6 and used a smaller learning rate to ensure that the difference between the training and testing R^2 values remained small (e.g. 0.452 vs 0.342 for XGBoost) as opposed to being large due to overfitting from an unconstrained forest.

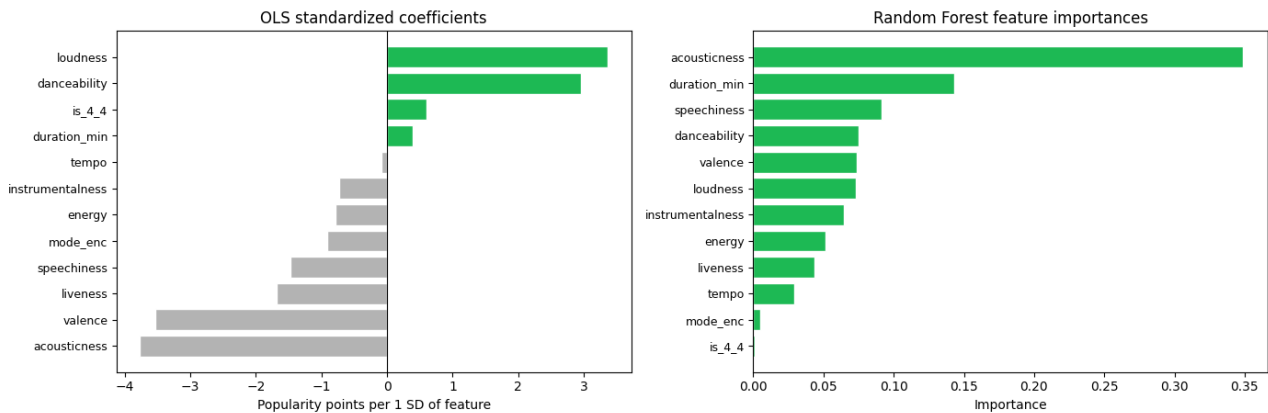
Table 3: Regression model comparison. Baseline (predict-mean) test RMSE = 17.42, $R^2 \approx 0$.

Model	Train RMSE	Test RMSE	Train R^2	Test R^2	Hyperparameters
OLS	15.51	15.57	0.204	0.201	—
Ridge	15.51	15.57	0.204	0.201	alpha = 50
Lasso	15.51	15.57	0.204	0.201	alpha = 0.001
Random Forest	12.89	14.16	0.450	0.339	depth 20, min_leaf 30, 200 trees
XGBoost	12.88	14.13	0.452	0.342	depth 6, lr 0.1, 400 trees

Figure 6 examines what drives the predictions, comparing standardized OLS coefficients with Random Forest feature importances (the two best models, Random Forest and XGBoost, are statistically tied and give near-identical importance rankings). Two points stand out. First, valence exhibits a classic suppression effect: although its marginal correlation with popularity is weakly positive ($r = +0.05$), its OLS coefficient is strongly negative (-3.5), second only to acousticness. Once the more danceable and louder tracks it tends to accompany are held constant, higher valence is associated with lower popularity. Second, the two model families disagree about duration: duration_min has a near-zero linear coefficient but ranks among the most important features for the Random Forest. This is exactly the kind of non-linear, threshold-like effect (very short or very long tracks being systematically less popular) that a single linear slope cannot express but a tree ensemble can, and it explains part of the ensembles' advantage. Acousticness is the strongest negative predictor in both views, and loudness and danceability the strongest positive.

Taken together, the results show a two-level ceiling. Non-linear models clearly beat linear ones, so the linear R^2 of 0.20 was partly a model limitation. But even two independent non-linear families converge at a test R^2 of about 0.34, extracting only about a third of the variance in popularity. This is the genuine ceiling of the feature set: the remaining two-thirds of the variance is driven by factors the audio features do not observe, such as a track's recency, its artist's prominence, marketing, and playlist placement. Audio characteristics are therefore a real but partial determinant of popularity.

Figure 6: Feature importance for popularity (OLS coefficients vs. Random Forest)



4.2 Classification

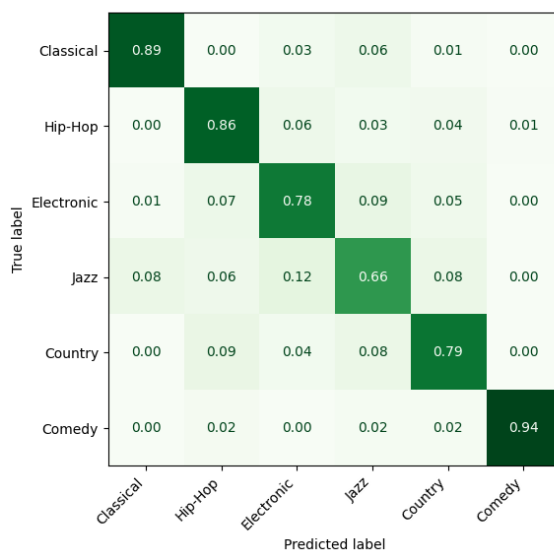
Table 4 compares the four classifiers. All four vastly exceed the 0.179 majority-class baseline, confirming that genre is strongly expressed in the audio features. Unlike the regression task, the models separate clearly. The Random Forest is the best by a wide margin (test accuracy 0.824, macro-F1 0.820), ahead of multinomial logistic regression (0.764), LDA (0.749), and QDA (0.734). The ordering of the linear models matches the cross-validation analysis of Section 2.4 exactly, and LDA's held-out accuracy (0.749) is essentially identical to its cross-validated estimate (0.743), indicating no over-fitting. That the Random Forest opens a roughly six-point gap over the best linear model shows that genre boundaries in feature space are meaningfully non-linear.

Table 4: Genre classifier comparison. Majority-class baseline accuracy = 0.179.

Model	Accuracy	Macro P	Macro R	Macro F1	Hyperparameters
Logistic Regression	0.764	0.760	0.760	0.756	C = 10
LDA	0.749	0.746	0.744	0.742	—
QDA	0.734	0.733	0.729	0.719	—
Random Forest	0.824	0.821	0.820	0.820	Depth = 20; min_leaf = 5; trees = 200

The row-normalized confusion matrix in Figure 7 provides confirmation of the predictions that could be drawn from the genre signatures of Section 2.4. Comedy is predicted to have very high classification accuracy (recall = 0.94, precision = 0.99) due to its extremely high levels of speechiness and liveness. Similarly, classical music (recall = 0.89) and hip-hop (recall = 0.86) appear to be clearly distinguishable by this model. However, jazz was found to be the most difficult genre for this model to classify (recall = 0.66). This is consistent with our prediction since jazz was the only genre among those studied that lacked a strong discriminating signature. Importantly, jazz misclassifications do not seem to be random. They occur primarily when jazz is confused with electronic music (0.12), classical music (0.08), and country music (0.08), all of which share substantial overlap with jazz in terms of their acoustic characteristics. Thus, the classification errors of this model seem to be musically interpretable rather than arbitrary.

Figure 7: Row-normalized confusion matrix



5. Conclusion

This study applied a set of twelve audio features to two contrasting supervised tasks. For regression, popularity proved only partially predictable: linear models explained about a fifth of the variance and flexible non-linear ensembles (Random Forest and XGBoost) converged at a test R^2 of roughly 0.34. Because two independent model families reach the same ceiling, this is best read as a genuine limit of the feature set; popularity is substantially driven by factors the audio features do not capture, such as recency, artist prominence, marketing, and playlist placement, rather than a shortcoming of any one model. For classification, the same features identified a track's genre with 82% accuracy, more than four times the majority-class baseline, and the model's errors were musically coherent, concentrated where genres genuinely overlap in acoustic space.

That the identical features are weak predictors of popularity yet strong predictors of genre illustrates how decisively the choice of target, not the modelling machinery, governs what is achievable when the inputs are fixed. This lesson was reinforced by the abandoned initial target, musical mode, which the same features could not predict at all; documenting why mode failed guided the move to genre.

Several limitations qualify these results. The regression ceiling reflects unobserved drivers of popularity rather than model capacity, so richer data (release date, artist metadata, playlist features) would be needed to push it higher. For classification, restricting to six well-separated genres makes the task cleaner than full 27-genre classification would be, and 1.5% of candidate tracks that spanned more than one of the six were dropped, mildly idealizing the label set. Finally, the tree-based models used fixed, sensible hyperparameters rather than exhaustively cross-validated ones; given how flat performance was across settings, a fuller search would be unlikely to change the qualitative conclusions.

6. References

Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

Hamidani, Z. (n.d.) *Ultimate Spotify Tracks DB* [Dataset]. Kaggle. Available at: <https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks-db> (Accessed: 8 July 2026).

Kalogeropoulos, K. (2026) *ME315 Machine Learning in Practice: Official Lecture Slides, Unofficial Lecture Slides, & Class Recordings, Days 1–9*. LSE Summer School.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, E. (2011) 'Scikit-learn: Machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.